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(54) **SYSTEM PROVIDING SELF-SERVICE
ACCESS TO LOCKED MERCHANDISE**

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(57)

ABSTRACT

A system providing self-service access to locked merchandise comprising: (a) providing a fixture that restricts access to the locked merchandise, wherein the fixture can automatically lock or unlock, allowing or restricting access to the locked merchandise; (b) providing a means of uniquely identifying an individual attempting to access the merchandise; (c) measuring a set of behaviors of the individual during any time the fixture is an open mode; (d) assessing whether the set of behaviors of the individual are suspicious or not relative to a set of suspicious event thresholds; (e) storing the individual and their set of behaviors as accessible records in at least one database; and (f) providing an algorithm which determines future access privileges of the individual to the enclosure based on a set of variables.

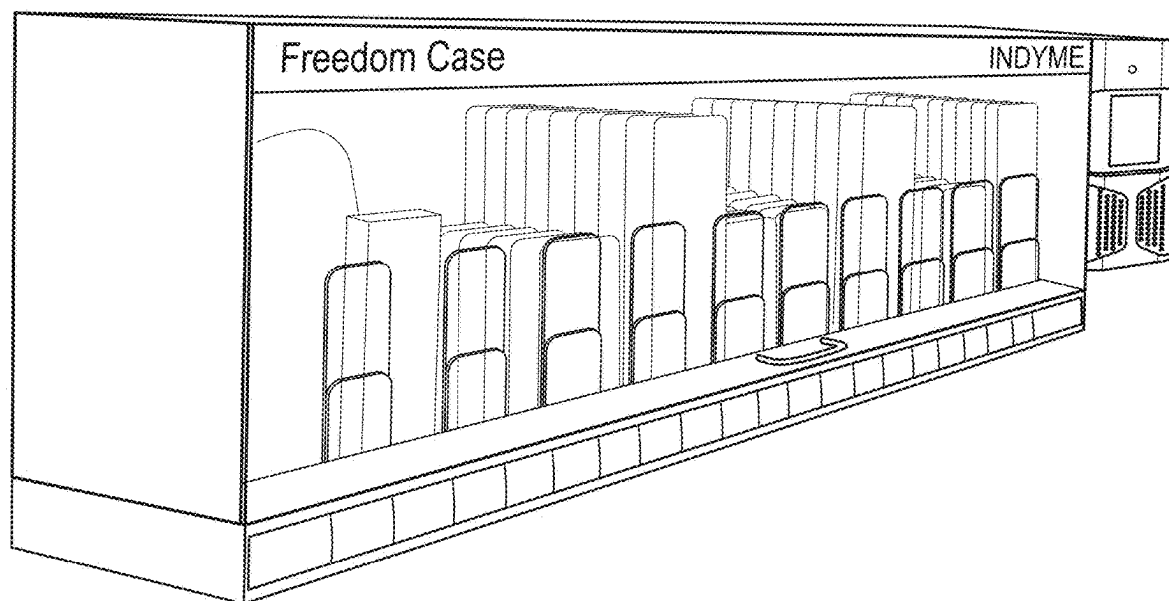


FIG. 1

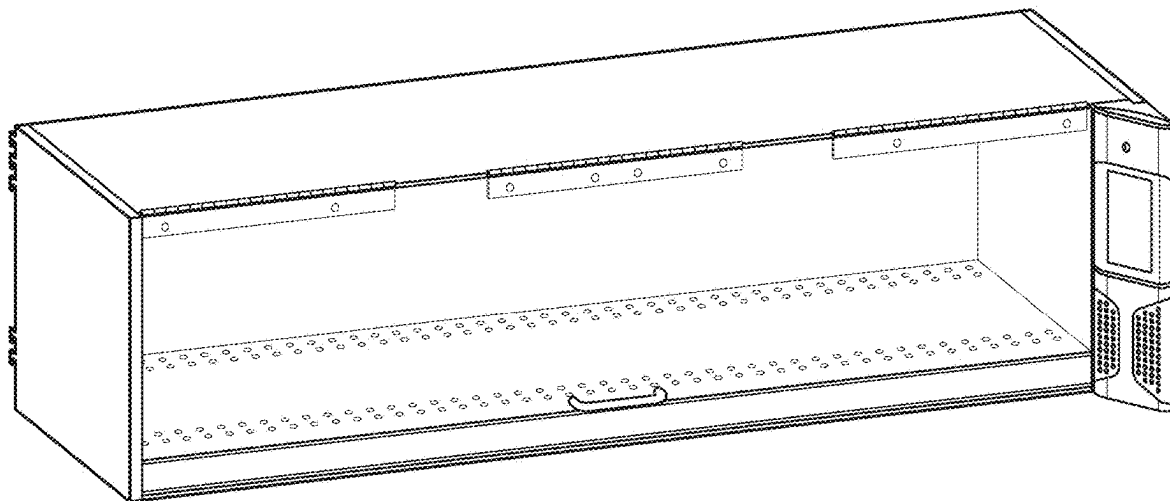


FIG. 2

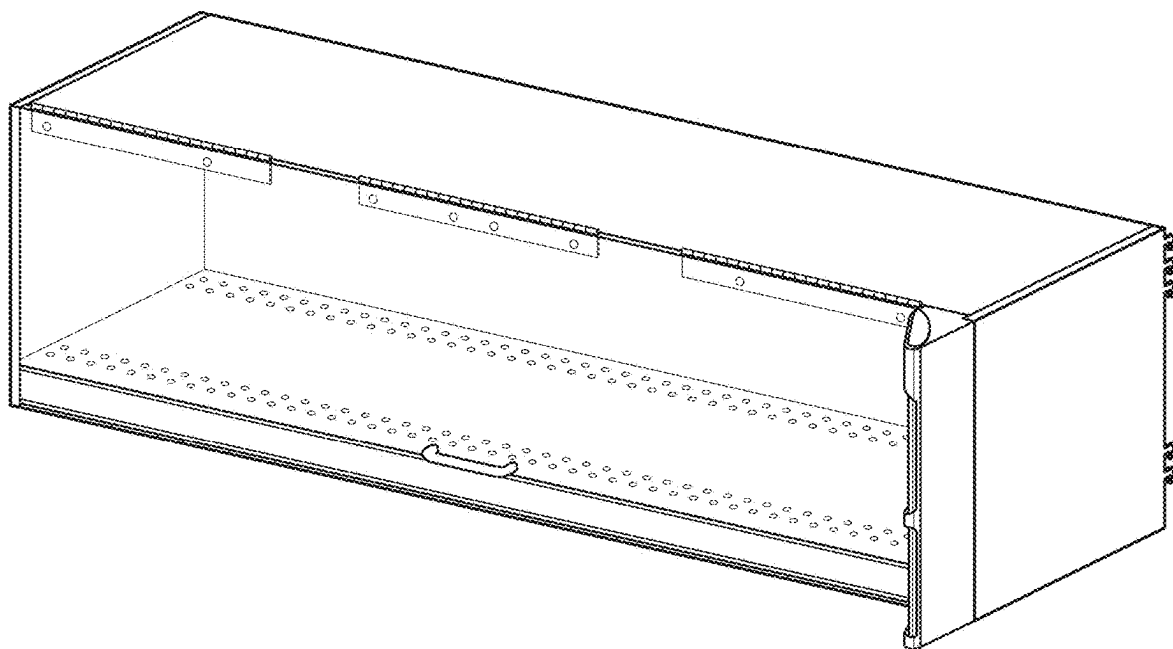


FIG. 3



FIG. 4

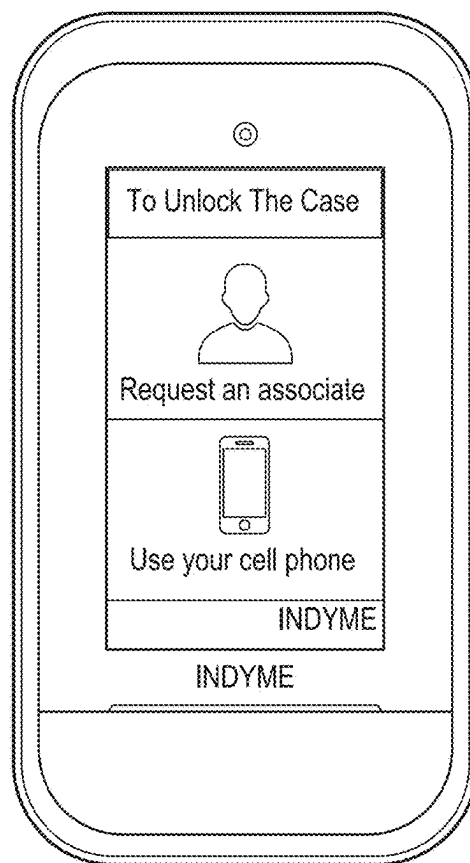


FIG. 5

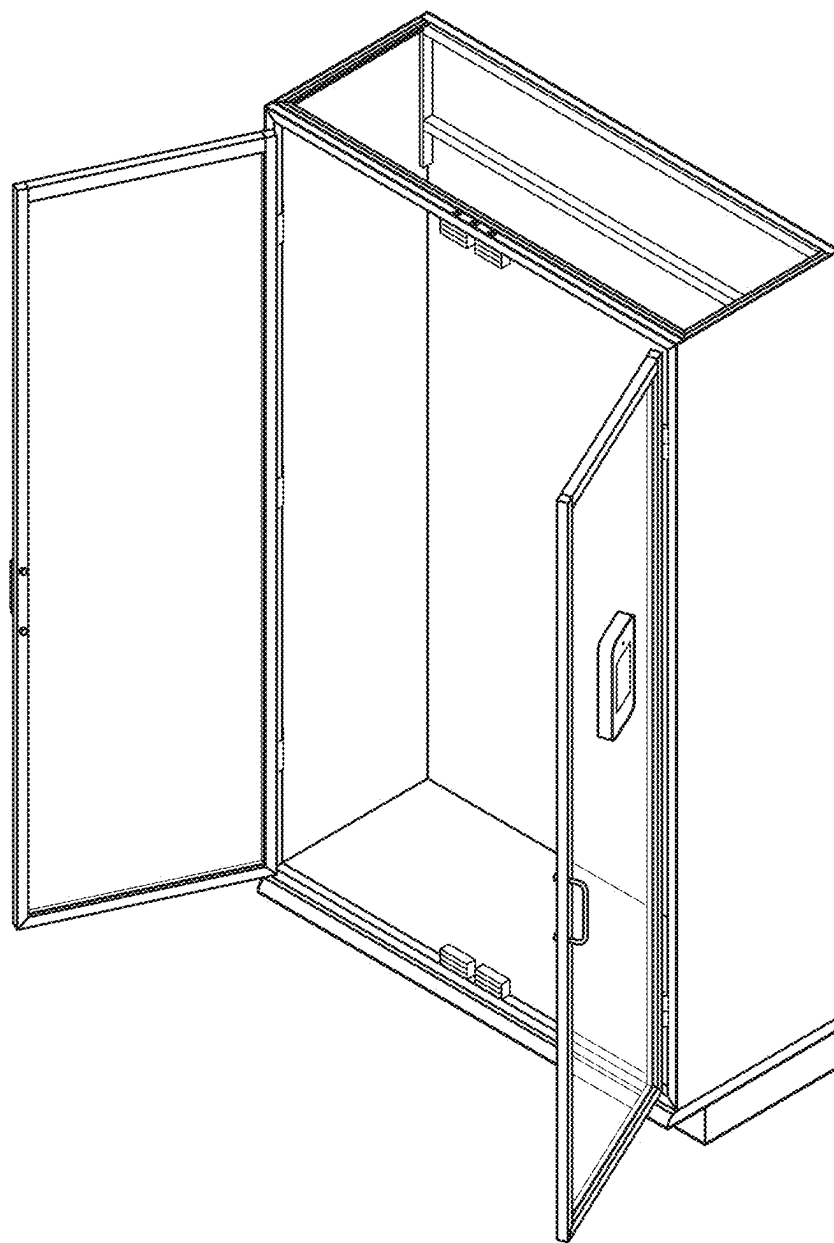


FIG. 6

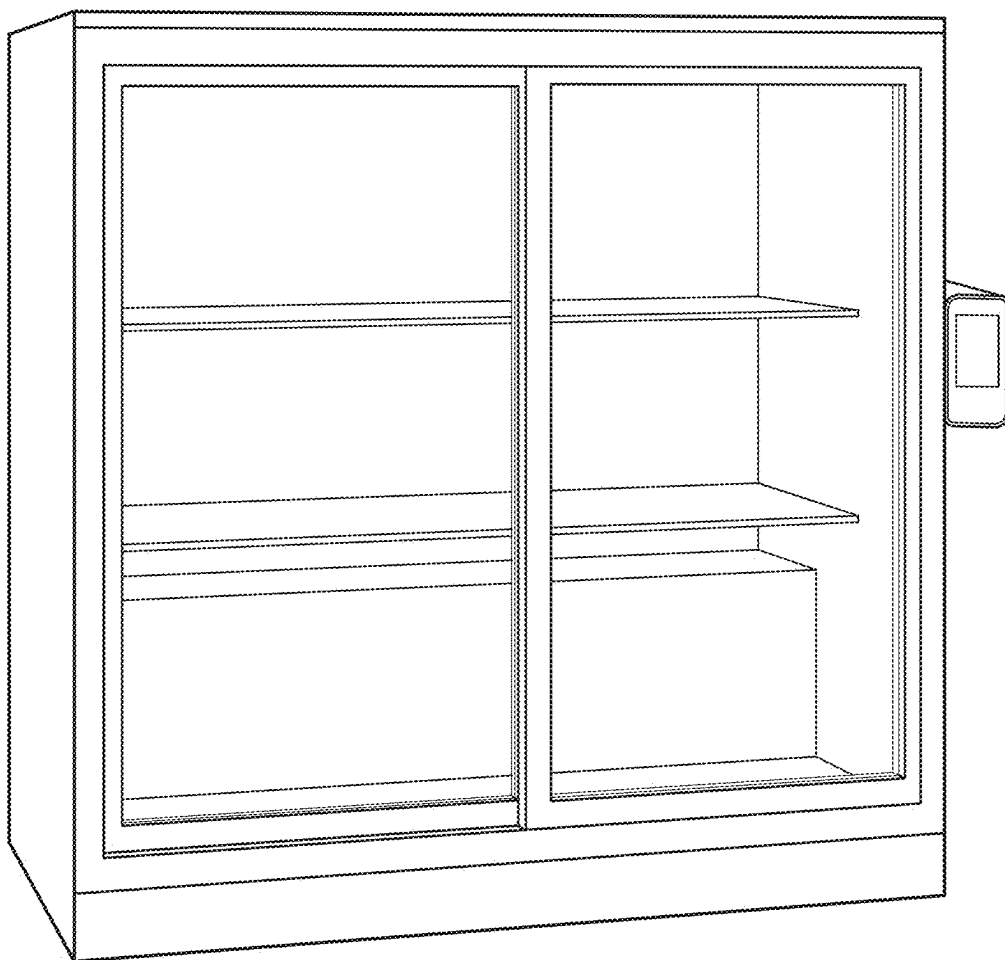


FIG. 7

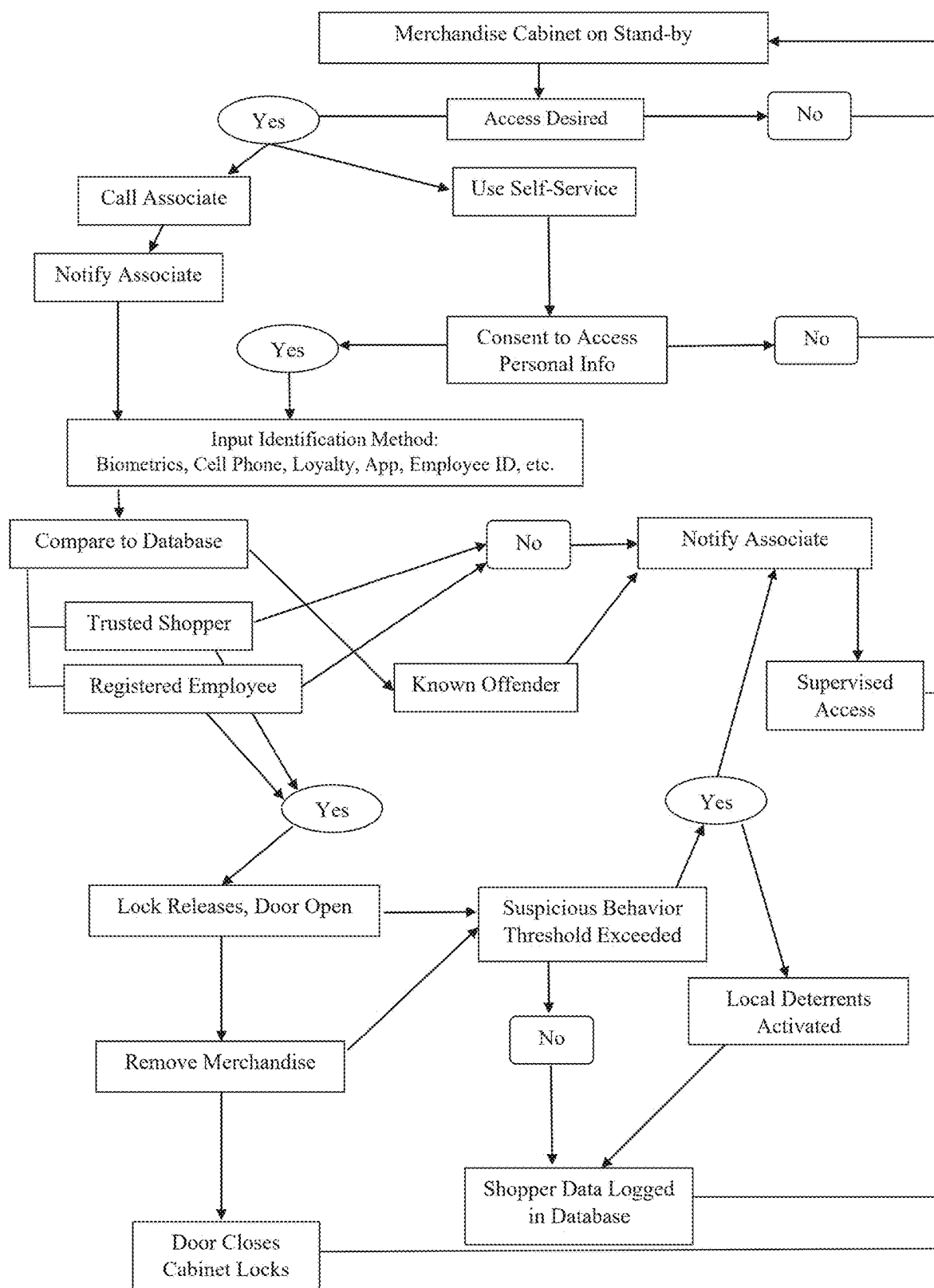


FIG. 8

SYSTEM PROVIDING SELF-SERVICE ACCESS TO LOCKED MERCHANDISE

RELATED APPLICATIONS

[0001] This application is a Continuation of U.S. application Ser. No. 18/602,514 filed Mar. 12, 2024, now U.S. Pat. No. 12,271,910, which is a Continuation of U.S. application Ser. No. 18/097,992 filed Jan. 17, 2023, now U.S. Pat. No. 11,928,694, which is a Continuation of U.S. application Ser. No. 17/737,404 filed May 5, 2022, which is a Continuation of U.S. application Ser. No. 17/567,765 filed Jan. 3, 2022, which is a continuation of U.S. application Ser. No. 17/197,951 filed Mar. 10, 2021, now U.S. Pat. No. 11,216,827, which is a continuation of U.S. application Ser. No. 16/940,168 filed Jul. 27, 2020, now U.S. Pat. No. 11,182,803, which claims the benefit of priority from U.S. Provisional Patent Application No. 62/878,747 filed Jul. 26, 2019, the entire contents of which are herein incorporated by reference.

BACKGROUND OF THE INVENTION

[0002] There is a universal paradox in retail stores, the better you protect your merchandise from theft, the greater the impact this protection has on product sales. In their effort to reduce theft of merchandise, retailers place restrictions on access to merchandise, impacting the vast majority of legitimate shoppers. The typical outcome is retailers inconvenience 99% of legitimate shoppers to stop theft from the 1% who come into their stores to steal. For example, a common method of reducing theft in a retail environment is to secure high-value merchandise by locking it up in a cabinet or other limited access merchandise fixture that makes it difficult for legitimate shoppers to access the product. To complete a purchase, the shopper must locate a store associate to unlock the fixture, gain access to the merchandise, and ultimately purchase the desired item(s). This process is not only costly and labor intensive for retailers, but it's time consuming, frustrating, and inconvenient, for the shopper, which nearly always leads to a horrible customer experience. This locked fixture approach has existed for year, virtually unchanged, and remains a significant problem for retailers. The primary goal of this invention is to modernize this outdated approach and to provide retailers with a means of protecting merchandise from theft, while creating virtually open access to this high value merchandise by legitimate shoppers, arguably providing retailers with the best of both worlds.

[0003] The recent trend of people willing to trade privacy for convenience is advantageous to the invention. More and more people willingly surrendering personal information to more easily access their phones, ATM's, board airplanes and other services. This invention capitalizes on this trend by requiring shoppers to provide personal identifying information in exchange for the convenience of unfettered access to merchandise. A typical example would be a shopper desiring to purchase high-value merchandise secured inside a locked liquor cabinet. Without this invention, the shopper would have to track down a store associate who has to locate the key to open the cabinet and access the goods. This cumbersome, time consuming, and inconvenient process typically results in a loss of 25% to 50% of sales; all this to thwart theft from 1% of the shoppers. Using this invention, a shopper approaching the same locked liquor cabinet is identified by some means, such as facial recognition, the use of the retailer's app or some form of customer loyalty

information, and the cabinet automatically unlocks. So long as this same shopper doesn't exhibit suspicious behavior, as described in this disclosure, the cabinet automatically opens each time this shopper returns.

[0004] While locking or otherwise securing merchandise reduces or even eliminates theft, it also suppresses legitimate sales due to the inconvenience of or even inability of the shopper to obtain required assistance when and where needed. This inconvenience causes a significant percentage of shoppers to abandon the purchase. The value of these lost sales often exceeds the savings realized by preventing theft. Even worse, shoppers significantly inconvenienced in a store often adversely impacts loyalty (i.e., a regular shopper may choose to switch to a competitive store for future shopping trips), representing a far larger loss than the missed sale of the protected item.

[0005] Given this tension between theft and sales losses, retailers often choose sales preservation over loss prevention. It is simply more financially advantageous to suffer theft losses rather than sales losses caused by securing theft-prone merchandise. This invention fundamentally eliminates this tension. It allows product protection from theft even while enabling free access to merchandise by trusted customers. In short, retailers no longer have to choose between protecting products from theft and reducing sales. Instead, legitimate shoppers are provided ready access to protected merchandise while such automatic access is denied to individuals deemed "untrusted" by the retailer. The self-service system described in this invention enables retailers to implement effective loss prevention measures which no longer adversely impacts merchandise sales or the shopping experience. Further, the invention supports loss prevention measures that mitigate both opportunistic shoplifting (typically an individual stealing an item) as well as multi-item sweeps (theft of many items at the same time) typically committed by booster teams as part of large-scale organized retail crime operations.

[0006] While the above antitheft scenario inspired the invention, it can also be used advantageously in other situations in which identified individuals that exhibit desired behaviors are rewarded with unfettered access to merchandise or other privileges.

[0007] In addition to the sales recovery and customer experience advantages provided by the invention, there are significant labor savings achieved by allowing customers to self-service locked merchandise. A typical transaction requiring a store associate to respond to a customer request to unlock a case can require up to 10 minutes of "task interruption" time for the store associate. From responding to the customer, locating the key, unlocking the cabinet and allowing the customer to shop the case, and then returning to tasking, this often takes 10 minutes. 10 minutes of store associate time is equivalent to ~\$2.50. This cost along with the lost productivity can easily eliminate any profit from merchandise sold in the locked fixture.

[0008] Finally, lost keys and re-keying of locked cases is a real problem in retail. Employees misplace keys, they bring them home after their shift, and the mechanism eventually wears out. This not only causes lost sales due to then inability to access the merchandise, but the cost of re-keying fixtures is significant.

SUMMARY OF THE INVENTION

[0009] The present invention provides for a system providing self-service access to locked merchandise comprising: (a) providing a fixture that restricts access to the locked merchandise, wherein the fixture can automatically lock or unlock, allowing or restricting access to the locked merchandise; (b) providing a means of uniquely identifying an individual attempting to access the merchandise; (c) measuring a set of behaviors of the individual during any time the fixture is an open mode; (d) assessing whether the set of behaviors of the individual are suspicious or not relative to a set of suspicious event thresholds; (e) storing the individual and their set of behaviors as accessible records in at least one database; and (f) providing an algorithm which determines future access privileges of the individual to the enclosure based on a set of variables. Preferably, the fixture admits or restricts access based on a trusted shopper score assigned to the individual when compared to a trusted shopper score threshold. Most preferably, the trusted shopper score increases when the individual exhibits normal behaviors and the trusted shopper score decreases when the individual exhibits suspicious behaviors. Optionally, at least one database is selected from the group consisting of a customer, a VIP customer, a known offender, a banned customer, a store associate, merchandise vendor, security personnel, and other. Preferably, the system assigns the individual's behavior as either normal or suspicious by comparing the set of behaviors of the individuals relative to interactions with the merchandise contained in the fixture or their interactions with the fixture itself against the set of suspicious event thresholds. Most preferably, the system is capable of deploying a set of real time deterrents once a suspicious behavior threshold is crossed.

[0010] In another aspect, the means of uniquely identifying the individual is at least one selected from the group consisting of biometric identification methods, RFID, NFC, bar codes, QR codes, user ID and Passwords, credit or bank cards, driver's licenses, smart phone App's, and cell phones. Alternatively, the fixture restricts access to the merchandise in the fixture through restricting access to the merchandise relative to the individual until access is granted. Preferably, the system further comprises at least one sensor to track and monitor the set of behaviors from the individual.

[0011] In yet another aspect, the individual can be associated with a classification of at least one selected from the group consisting of a customer, a VIP customer, a known offender, a banned customer, a store associate, merchandise vendor, security personnel, and other. Preferably, the suspicious event threshold can be determined based on the classification of the individual. Alternatively, individuals classified as authorized store associates have suspicious event and trusted shopper score thresholds set to allow stocking of the fixture without such behavior being classified a suspicious behavior. Preferably, if a request to open the fixture is made by certain high risk classes of individuals, then the system will transmit a class specific response comprising sending a unique identifier (UID) to a store associate. Optionally, a class of individual can be denied access for a specified period of time. Preferably, the suspicious behavior threshold can be automatically set based on a statistical analysis of past behaviors from a number of individuals. Alternatively, a local real-time deterrent is deployed in real time when the suspicious event threshold is crossed. More preferably, the system will transmit a class

specific response comprising sending a unique identifier (UID) to a store associate if a request to open the fixture is made by certain high risk classes of individuals. Optionally, a user of the system can enroll known offenders, store associates, vendor personnel, and security personnel into its individual classification database.

[0012] The primary purpose of the invention is to maximize sales while minimizing theft. It also maximizes convenience and improves the shopping experience for the vast majority of legitimate shoppers while selectively denying unsupervised merchandise access by likely shoplifters. The invention intelligently enables and disables real time deterrents based on a given shopper's past behaviors and other variables. By uniquely identifying each shopper at a merchandise display fixture and simultaneously observing and recording their behavior, the invention can determine if future product access will be granted to that shopper. Once a shopper exhibits suspicious behavior, anti-theft real time deterrents are deployed during the event, and also during subsequent visits. For example, a legitimate shopper will be granted free access to a locked merchandise cabinet while an "untrusted" individual is denied unsupervised access. This is accomplished by uniquely identifying the person (though not necessarily "by name") and determining if past behavior or other factors warrant granting that person access to the merchandise. Once access is granted, the invention may trigger local alarms or real time store personnel notification if the shopper exhibits suspicious behaviors resembling theft events in progress. The invention draws upon several methods for uniquely identifying shoppers requesting access to protected merchandise, and several other methods for identifying behaviors deemed normal, undesirable, or suspicious. By coupling the elements of personal identification, product movement sensing technologies, database correlations, and algorithms, access to protected merchandise can be optimally managed.

[0013] In certain configurations, the restricted access fixture may contain multiple locked fixtures which are coupled to a single customer facing User Interface and or local controller. This centralized controller provides a single point of individual authentication and facilitates access to the all the associated locked fixtures. Using one method of accessing the multitude of locked cases, the central authentication controller would authenticate the individual requesting access to the locked fixtures as a trusted shopper, once authenticated, all locked fixtures coupled to that controller would open. Once the shopper opens one of the coupled fixtures, the other locked fixtures would re-lock. In another embodiment, the user can select a specific locked fixture of the multiple locked fixtures to access, and only that specific fixture would be opened. For all these multi-locked fixtures use cases, the normal authentication processes, suspicious event detection processes, deterrent activations, employee notifications etc. which are utilized in a single locked fixture configuration would be utilized in this multi-case configuration.

[0014] It should be noted that the self-service authentication methods used to access to locked merchandise practiced by this invention are optional and are accessed on an "opt-in basis". Any customer who is uncomfortable utilizing these identification and authentication methodologies or is uncomfortable trading personal identifying information in exchange for self service access to locked merchandise may

simply summon a store associate through the system and the store associate can provide access to the locked merchandise.

[0015] This invention has other applications where past behaviors can be monitored to grant or deny privileges in the future. Since the invention correlates uniquely identified individuals with merchandise interaction behaviors, it can also be utilized at fixed self-checkout stations as well as self-checkout via mobile devices to identify “trusted shoppers” who have exhibited repeated good behaviors as well as “untrusted shoppers” who have exhibited suspicious behaviors in the past.

[0016] While reasonable variations of the operating sequence may be implemented per the detailed description, the following represents a typical operating sequence of the invention.

[0017] First, a person desiring access to the locked merchandise fixture encounters a visual display adjacent to the locked fixture which presents the customer with two options, to either call an associate to unlock the fixture in the traditional way, or to use personal identification to enable self-service access to the locked fixture. The customer so indicates their choice by pressing a soft button on the touch screen display.

[0018] If the customer selects the self-service option, they are then presented a disclosure statement which provides “informed consent” for the system to utilize personal information for the purpose of unlocking the fixture. If the system is using biometrics such as facial recognition to uniquely identify the shopper, a visual and/or audible message then instructs the user to look directly at the camera typically located above the display. The display presents the customer with an image of the shopper from the embedded camera. The customer aligns their face with a box on the screen, thus providing a perfect “mug shot” of the individual. The system then compares the biometric information with a database to determine if this individual is a “trusted shopper” and will be allowed access to the locked case.

[0019] Once the customer is determined to be a trusted shopper, the lock releases and the display indicates the fixture is open. The customer opens the case and removes the desired merchandise. The system counts the units removed from the fixture while open, and the case automatically locks once the fixture is closed. The customer’s biometric information and merchandise removal behavior are data logged in the system database.

[0020] In the event the shopper exhibits a set of behaviors that exceeds a per-determined suspicious event threshold, the system can deploy local real-time deterrents and/or notify store associates.

[0021] If the shopper is a known offender or is a shopper who has exhibited repeated suspicious behaviors in the past, they will be denied self-service access to locked merchandise and the system notifies a store associate to open the locked fixture and provide “supervised access” to the locked merchandise.

[0022] In the case of denial of self-service access, a store associate is notified and once enrolled in the system, the store associate can use their own personal identifying information to unlock the case without the need for keys.

[0023] There are multiple methodologies for achieving the operating sequence described above and the detailed

description that follows illustrates the various methodologies and sensor which can be used to achieve this operational sequence.

BRIEF DESCRIPTION OF DRAWINGS

[0024] The novel features of this invention, as well as the invention itself, both as to its structure and its operation, will be best understood from the accompanying drawings, taken in conjunction with the accompanying description, in which:

[0025] FIG. 1 illustrates an exemplary embodiment of a fixture with user interface of the present invention;

[0026] FIG. 2 illustrates an alternative exemplary embodiment of a fixture with user interface of the present invention;

[0027] FIG. 3 illustrates an alternative view of the fixture of FIG. 2;

[0028] FIG. 4 illustrates an alternative exemplary embodiment of a fixture with user interface of the present invention;

[0029] FIG. 5 illustrates an exemplary user interface of the present invention;

[0030] FIG. 6 illustrates an alternative exemplary embodiment of a fixture with user interface of the present invention;

[0031] FIG. 7 illustrates an alternative exemplary embodiment of a fixture with user interface of the present invention; and

[0032] FIG. 8 illustrates an overview of the self-access system present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0033] The invention is best understood by first gaining insight into the types of sensors and devices which are coupled together to enable its operation. Keeping in mind that a specific attribute of the invention is that is the operation of the invention is not dependent on any one specific sensing or merchandise access restriction technology. Three categories of sensors and devices are typically integrated to operate the invention:

[0034] People identification sensors capable of uniquely identifying an individual;

[0035] Merchandise interaction (customer behaviors) sensors.

[0036] Real-Time Deterrents Merchandise protection and suspicious behavior notification devices;

[0037] Using the information and controllable access provided by these device categories, the invention determines if a given person is considered “trusted” and will be granted merchandise access. A more detailed description of each category follows.

Person Identification Sensors

[0038] The invention requires a method of uniquely and repeatedly identifying each person for which the invention is considering granting merchandise access. As these sessions may occur on different days, at different locations within the store, and even at different stores the identification method must remain accurate even when an individual wears different clothes, hair styles, etc. The invention also identifies store associates and allows them, once registered, access to merchandise for stocking purposes. The invention’s core functionality does not require identifying the person “by name,” though some ancillary benefits, described later, may be realized with this functionality.

[0039] Any method of identification meeting the above requirements are acceptable. These include but are not limited to:

- [0040]** Facial recognition using video cameras;
- [0041]** Other biometric methods, such as iris recognition, fingerprint detection, or any other biometric detection that can uniquely identify an individual;
- [0042]** RFID, Bar Code, QR Code, and other methods integral to a card or other device issued by the store (such as a loyalty card selectively provided to customers by the store or even a driver's license);
- [0043]** Cell phones carried by shoppers which may use local connectivity (e.g., WiFi, Bluetooth and/or NFC) as a means of identification;
- [0044]** Smart phones carried by shoppers that are logged in to the retailer's app (the login considered adequate proof of identification);
- [0045]** Using onboard biometric identification capabilities on smart phones carried by shoppers, then relaying this confirmation to the invention via the retailer's App, by presenting the shopper's phone screen (displaying a QR or similar code) to the invention's camera, or via a text message; the transaction could be initiated on the shopper's phone resulting in identification confirmation to the invention or the shopper could scan a QR or similar code displayed by the invention to automatically launch the identification process and the installed retailer's App on the shopper's phone;
- [0046]** Smart phones carried by shoppers accessing a web site that performs biometric identification; this method does not require an installed retailer's App on the shopper's phone and could be automatically launched by scanning a QR or similar code at the fixture.
- [0047]** Any other device or method issued by the retailer which uniquely identifies the individual

Location Confirmation of Identified Persons

[0048] Some methods of person identification inherently confirm the physical presence of the person at the locked fixture. For example, facial recognition by a camera at the fixture. However, some identification methods described above using the shopper's smart phone do not provide this assurance, making it possible to intentionally or accidentally unlock a fixture from a remote location, which is undesirable. Methods to confirm physical location in such situations include but are not limited to:

- [0049]** Short Range Wireless Connectivity with Shopper's Smart Phone: Typically using Near Field Communication (NFC) or Bluetooth, both of which have limited range but NFC being the most suitable;
- [0050]** Location Based Services Detecting Shopper's Smart Phone: WiFi services providing "triangulation" methods;
- [0051]** Dynamic QR Codes: While a shopper's phone could scan a QR Code label affixed at the fixture location, a photo of that QR Code could be used at any time and any place to inappropriately trigger an unlock command. To avoid this, Dynamic QR Codes displayed by the invention, or a nearby discrete device, change periodically or even with every transaction.

Merchandise Protection and Suspicious Behavior Notification Devices

[0052] The invention is typically used with one or more devices providing a means of securing and/or limiting access to merchandise. A requirement of these devices is a means by which control signals from the invention may allow or deny complete and/or limited access to the protected merchandise. The possible types of devices that could meet this requirement are virtually unlimited; some examples:

- [0053]** Locked Cabinets/Doors: Typically, an electronically activated locking device grants or denies access to the protected merchandise behind the door. These range from fully enclosed cabinets to flip doors in front of merchandise shelves to refrigeration doors opened to access beer or other high-theft merchandise;
- [0054]** Dispensing Display Fixtures: A wide variety of devices fit this category which includes: peg hooks with a knobbed spiral coil that must be turned to permit removal of one item at a time; push button devices that dispense one item when pressed (much like a vending machine); merchandise "pushers" that keep a facing of merchandise pressed tightly against the front of a shelf; and specialized fixtures (such as for canned baby formula) that rely on a specific form factor for limiting the number of units dispensed. Whether the operating mechanism is largely mechanical or electronic, all compatible devices can be controlled by the invention to permit (or not) dispensing of one or more of the protected products or, in the case of pushers, to disable the pushing function to limit access to merchandise.
- [0055]** Movable Fence: Any kind of sliding fence that can be moved to gain access to the product.

Fixture Status Sensors

[0056] All of the fixtures the invention is compatible with would have sensor on the door/fence/dispensing mechanism which can detect the status of the fixture, namely, is the fixture "open" in the sense that merchandise can be accessed, or is it "closed" in the sense that access to the merchandise is restricted or is inaccessible in the closed state.

Real Time Deterrents

[0057] Anti-theft deterrent devices can be controlled by the invention including various communication, notification, and alarms devices. Using product removal information from internal sensors such as the merchandise movement sensors, fixture status sensor, camera tampering sensor etc., to detect suspicious events the invention can trigger local deterrents such as notifying store personnel and/or activating local deterrence devices which may be local alarms, flashing lights, audio such as PA announcements, turn on cameras, or any other traditional anti-theft deterrent device which would deter the offender from proceeding with the theft.

[0058] In one possible configuration, merchandise may not be physically protected from access but rather the invention notifies store personnel and/or activates local deterrence devices mentioned above when an "untrusted" person is detected interacting with or removing merchandise.

Customer Behavior Sensors Merchandise Interaction Sensors

[0059] It is desirable, though not always required, that sensors inform the invention of the characteristics of the customer's set of behaviors including merchandise interaction by the identified individual accessing the merchandise. This may include the following detections (and possible invention characterizations):

- [0060]** No merchandise removed (typically considered a non-event);
- [0061]** One item removed, possibly including identification of the specific merchandise item (a normal shopper purchase);
- [0062]** Multiple items removed within normal shopper patterns (likely not considered a suspicious event);
- [0063]** Rapid or other abnormal removal of multiple items (a suspicious event).
- [0064]** Multiple items removed of the same product beyond normal shopper patterns (likely a suspicious event)
- [0065]** Replacement of product, as in a merchandise replenishment or stocking action by a store associate or simply a customer changing their mind about a purchase.

[0066] The invention is not dependent on any specific method of detecting these interactions and it is to be understood that not all acceptable methods of detection are capable of providing all of the interaction characterizations listed above. The type of merchandise interaction sensor is typically determined and/or limited by the merchandise protection method used (e.g., locking case, peg hook, etc.). Generally, any sensor capable of providing at least some insight into the nature of merchandise interaction can be used by the invention. These sensors have the common capability of measuring product interaction and some have the additional capability of measuring inventory and specific product SKU level information. These sensing technologies include, but are not limited to, the following:

- [0067]** Cameras (including 3D cameras) with integral or post-processing capabilities able to detect merchandise interactions which may include detection of hand penetrations into an open fixture and quantity of items held in a hand during removal; these cameras can also count the number of units before and after a transaction, thereby "netting" the number of items removed less the number replaced. These cameras could also be equipped with Artificial Intelligence capable of detecting the type of merchandise contained in the fixture down to the SKU level. This SKU data is a useful input to both the TSS algorithm and for stocking and replenishment purposes.
- [0068]** Vibration sensors able to detect the movement of merchandise on/off shelves, peg hooks, and other displays;
- [0069]** Switches (including magnetic switches and similar) detecting merchandise removal on dispensing display fixtures.
- [0070]** Weight measuring pads that can determine the number of items on a shelf.
- [0071]** Light detecting devices which can determine the location and number of items on a shelf.
- [0072]** RFID tags which can be detected by a receiver and count units being removed from a fixture or being replaced. This technology can also uniquely identify

the product down to the SKU level. This SKU data is a useful input to both the TSS algorithm and for stocking and replenishment purposes.

- [0073]** Smart Fixtures: any fixture with sensing technology that can determine units being removed from a fixture and also the inventory position of each product contained in the fixture. This inventory data is a useful input for stocking and replenishment purposes.

- [0074]** Sound detection: fixtures which detect an audible sound related to product removal

[0075] As noted previously, not all of these methods can reliably provide all possible merchandise interaction characteristics. However, the invention can be configured to optimize for best use of those characterizations that can be provided.

The Trusted Shopper Algorithm and Development of the "Trusted Shopper Score"

[0076] An individual's privileged access to secured merchandise is determined by an algorithm which takes as its primary input this individual's shopping behavior during each fixture access event. Based on the individual identification methodology utilized by the invention, the algorithm may have access to databases containing multiple additional attributes (ie. metadata) related to this individual which enable to the algorithm to consider these additional attributes to make more informed merchandise access decisions. These additional individual attributes are summarized in the section below entitled Shopper Metadata. This Metadata is used to place individuals into a "classification".

[0077] Past shopping behaviors, combined with these Metadata attributes resulting in an individual's classification are inputs to the algorithm and are used to develop a Trusted Shopper Score or TSS. This TSS is a numerical score which can be compared to a programmable Trusted Shopper Score threshold set by the retailer as the basis for ongoing access to locked merchandise. In fact, as different merchandise carries different risks of theft, the retailer could program each individual locked fixture with its own unique TSS threshold based on the merchandise contained in that fixture.

Past Shopping Behavior and Trusted Shopper Score

[0078] Each time the individual demonstrates trustworthy behavior during a fixture access event, the algorithm increases that person's TSS. Likewise, detected undesirable behavior (such as abnormal merchandise interaction activity and permitting "tailgating" or allowing other shoppers access to an open fixture) results in the reduction of that individual's Trusted Shopper Score. Once established, the Trusted Shopper Score is compared to a TSS threshold established by the retailer and as long as the individual remains above the threshold, future access privileges are maintained. If an individual's TSS drops below the threshold, the system will deny unsupervised access to the merchandise and will summon a store associate. For example, an aggressive TSS threshold might deny unsupervised access based on one suspicious merchandise interaction. In this aggressive example, a single incidence of unacceptable behavior effectively revokes that shopper's privileged access rights.

[0079] There are many reasons a retailer may not elect such an aggressive threshold. After all, retailers do not want to alienate loyal shoppers and often are more concerned

about sales losses than theft losses. Since it is possible that acceptable behavior may occasionally be incorrectly perceived as unacceptable, a more lenient threshold could be warranted such that a complete loss of unsupervised access occurs only when a very serious and definitive breach of behavior is detected and/or when multiple suspicious events are detected or a pattern of unacceptable behavior has occurred.

[0080] Another input to the algorithm is simply the passage of time. For example, when a shopper removes a large number of units from a fixture, this behavior is generally viewed as suspicious and would negatively affect their TSS. This could result in denied unsupervised access. However, this behavior may have been instigated by an in-store sale or may simply be forward buying behavior based solely on a discounted price or promotion. If a shopper forward buys several packs of razor blades for example, the algorithm may drop their TSS sufficiently to be restricted from accessing any locked merchandise. After the passage of a specified period of time, the algorithm would increase their TSS to an acceptable level to allow self-service access again. The algorithm may also allow the same individual to be granted access to other merchandise while being restricted from accessing razor blades for a specified period of time.

[0081] Accordingly, the set of shopping behavior factors that may be considered by the algorithm include, but are not limited to:

[0082] Immediate past behavior event

[0083] Multiple past behavior events

[0084] Patterns of behavior (such as attempts to “game the system” like alternating acceptable behavior and theft behavior to gain tokens)

[0085] Accuracy of sensing technology: The ability to detect suspicious behaviors is based on the type of behavior detection sensors; for example, dispensing devices typically have a very high accuracy on the quantity and specific merchandise removed while vibration detection devices are less accurate; video sensing accuracy is often dependent on camera placement and merchandise type.

[0086] Value of merchandise subjected to unacceptable removal.

[0087] Specific SKUs removed.

[0088] Time between past and current attempts to access the merchandise.

[0089] Shopper history; purchasing history; registered membership in store programs; high store App utilization; and/or long and frequent loyalty would be considered more trusted;

[0090] The inclusion and/or weighting of the above factors are used either alone, or in combination with Shopper Metadata, and the individual’s classification by the algorithm and result in a “Trusted Shopper Score” for that individual. The retailer then sets a TSS threshold which is used by the invention in evaluating merchandise access requests to locked fixtures. This same TSS may be used by other store systems for granting levels of shopper permissions, such as unsupervised self-checkout, use of store portable scanning devices, and other privileged services, or virtually any service which otherwise would require supervision by store personnel.

First Locked Fixture Interaction

[0091] The configuration variables of the invention enable the retailer to determine how to initiate new shoppers into the system. For example, the algorithm could be configured such that all first-time users (except those in the pre-existing “Known Offender” database, described later) are considered trusted and provided privileged access. So long as acceptable behavior is observed on the initial and subsequent interactions, access privilege is granted.

[0092] An alternative implementation might require shoppers to register or “opt-in” to the system (e.g., via a loyal shopper program or other identification means described above) as a prerequisite to being granted privileged access. By opting into the system, the shopper would then be granted greater shopping convenience. In this case, identification of the individual could link to a known “by name” loyal shopper database.

Self-Calibration of Suspicious Event Declaration

[0093] As previously noted, if the quantity of items removed from a monitored fixture exceeds the suspicious event threshold, that transaction is deemed a “Suspicious Event” and the shopper associated with the action will have their TSS negatively impacted (depending on the incident and other factors comprising their Trusted Shopper Score). The removal of an extreme quantity of merchandise in a single transaction, however, would exceed an even higher suspicious event threshold defining a “Sweep Event”. Even an individual with a high Trusted Shopper Score would likely be severely penalized if associated with a Sweep Event.

[0094] An acceptable number of product removals will vary based the type of merchandise contained in the locked fixture. As the number of locked fixtures utilizing the invention grows, a single retailer may have thousands of cases utilizing this self-service invention. Manually determining, configuring, and maintaining appropriate event thresholds, especially the Suspicious Event threshold across thousands of locked fixtures across hundreds of stores, can become very time consuming and perhaps impractical to manage. The fine line between a reasonably normal shopping quantity and a somewhat excessive quantity can vary by merchandise assortment, store location, and even season of the year. The invention’s threshold self-calibration function automates this task by periodically adjusting the Suspicious Event threshold based on a statistical analysis of customer behaviors over time.

[0095] Self-calibration requires the retailer to establish an initial Suspicious Event Rate by estimating the percentage of transactions representing Suspicious Events. For example, the retailer may estimate (as is generally the case) that 98 of its shoppers are trusted and legitimate, and therefore 2% of its shoppers are potential offenders. This estimate represents the percent of fixture opening events that would likely be related to theft events. This percentage establishes the Suspicious Event Rate. After an initial statistically relevant number of fixture opening events, the system would then automatically adjust the Suspicious Events Threshold (defined as the number of unit removals determined to be suspicious) based on actual measured product removal behavior observed by previous shoppers according to the percent threshold set by the retailer. The invention’s self-calibration algorithm retains product removal data from each

opening event and uses that data to calculate a suspicious event threshold based on the suspicious event rate indicated by the retailer. In this example, the number of units removed by the top 2 percent of shoppers would be deemed Suspicious Events. The invention then periodically adjusts the threshold on a rolling basis to take into account possible merchandise changes, sale events, and other external factors that would affect the number of product removals from the locked fixture.

[0096] This self-calibration algorithm uses statistical methods to eliminate “outliers” representing non-standard behaviors which would otherwise skew the Suspicious Event Threshold or other system data in general. For example, if the top 2% of shopper merchandise interactions contain non-standard behaviors, the self-calibrating algorithm would determine if these interactions are non-standard outliers, and remove these outliers from the calculation of the suspicious event threshold. To illustrate this scenario, if one or more of the 2% of shoppers used to establish the suspicious event threshold were to take 500% more than the remainder of the top 2% of shoppers, this could skew the average for this group and could impact one of the methods for calculating the suspicious event threshold. Removing these outliers will create a more accurate suspicious event threshold based on more standard behaviors.

[0097] This self-calibrating algorithm may accept external metadata and individual classification as inputs to more specifically calibrate each individual fixture and dynamically change the suspicious event threshold according to changes in merchandise, pricing, or other external factors that may affect product removal behaviors. This meta data may include but is not limited to the following; SKU level information about what merchandise is contained in the locked case (no matter how this data is collected), RFID information, planogram information, pricing data, as well as promotional information, weather events, any natural disaster events, civil unrest, or any other metadata which is likely to affect product removal behaviors.

[0098] It is advantageous to eliminate known merchandise stocking data from the self-calibration algorithm. In the event the system has access to metadata and is aware of individual classifications such as store associates, vendors or other non-customers accessing the locked fixture and the associated re-stocking events, the self-calibrating algorithm would ignore product interaction data associated with non-customer-initiated re-stocking events (or customer assistance requests for denied or opt-out customers). In this way, the algorithm may more accurately determine normal product removal thresholds without the additional outliers created by stocking events or other non-customer generated product removal events.

The Role of Metadata Databases and integration with other store systems

[0099] The invention’s benefits can be enhanced through access to various Metadata such as the following which would be used to develop individual’s classifications and other purposes:

[0100] Registered Loyal Shoppers Database—as noted previously, this can be useful as a means of rewarding loyal customers by initiating privileged access. The shopper’s habits and loyalty may be considered by the Trusted Shopper algorithm in deeming certain actions acceptable.

[0101] Purchase history of a shopper—by accessing a shopper’s personal shopping history, the TSS algorithm may provide for a more lenient treatment of suspicious events committed by shoppers who spend over a threshold with a retailer. This allows for the avoidance of denying self-service access by a retailer’s highest spending customers.

[0102] Known Offender Database—includes known perpetrators such as convicted shoplifters, suspected booster gang members, and other individuals banned from privileged access. This database can be an integration of data provided by local law enforcement, through the retailer’s own case management system or other known offender databases. Known offenders are denied self-service access but may still access the merchandise by the system notifying a store associate to unlock the fixture. The system can notify the store associate and communicate that a known offender is requesting access to the locked merchandise, allowing the associate to follow alternate product protection protocol such as bringing the merchandise to the cashier rather than handing it to the offender. Further, in the case of unsecured but monitored merchandise, merchandise interaction by those shoppers identified as known offenders will result in immediate store associate notification actions as well as deployment of local deterrents, as described earlier. It is especially noteworthy that the known offender Database is not exclusive to a specific store location but is a composite from across multiple stores in that retailer’s chain and perhaps even shared information from other retailers.

[0103] Store Associates Database—includes the identity of all store employees and is used to provide ready access to secured merchandise to the associate in the course of assisting shoppers not granted self-service privileged access; further, since the act of re-stocking merchandise can be interpreted by some sensors as a theft occurrence, an authorized store associate would avoid this determination (for this reason, the system may also refer to the time clock system database to verify the store associate is on duty as a condition to ignoring what would otherwise be deemed a suspicious event). This database can be coupled to the retailers HRIS system which can automatically enroll and remove store associates as they are hired and terminated from the company. Additionally, this data would be used by the self-calibrating algorithm to ensure only customer product removal data is used in determining the suspicious event thresholds.

[0104] Vendor Employee Database—in some cases, retailers ask merchandise vendors to re-stock shelves with their products. In these cases, non-retailer employees may be granted access to the locked cases the same way store associates are granted access. Much like the Store Associates Database, this provides identity and authorization information (e.g., which merchandise locations) regarding vendors who may re-stock certain merchandise.

[0105] Video Management Systems: If video is used as a sensing technology or if the merchandise is monitored by in-store video surveillance, this video stream can be “bookmarked” by the system to enable auditing of

system performance as well as the identification and possible apprehension of offenders who exhibit suspicious behaviors.

[0106] Self-Checkout Auditing: Increasingly retailers are employing self-checkouts. These self-checkouts have auditing procedures which can identify individuals who fail to scan all items or otherwise cheat the system. If individuals are uniquely identified using this process, this database can be integrated into the TSS algorithm to make better access decisions. Alternatively, once a shopper exhibits suspicious behavior at the locked case, if that customer is also identified at the self-checkout, the attendant may be notified to assist that shopper and ensure the items removed were properly checked out.

[0107] SKU Level Product Data: Awareness of the products contained in the case is useful to the TSS algorithm to more precisely set suspicious event thresholds. It is also useful for stocking and replenishment purposes, as well as compliance with retailer policies for which products should be placed in the fixture.

[0108] Gender, Age, and Mood Data: When using biometric or other personal identification methodologies, these systems are capable of determining age, gender and even mood. These inputs are useful to the system in identifying customers who may need assistance. This information is also useful to the retailer for marketing and merchandising by identifying the demographics of the customers who are accessing certain SKUs. This information can be used to display more targeted advertisements on the display of the invention or other specific local advertising vehicles.

When Merchandise Access is Denied or Delayed

[0109] In the event a shopper's Trusted Shopper Score is below the TSS threshold and requested access is denied, the invention summons assistance to meet the customer's needs; methods include:

[0110] Customer Assistance Request: An audio or text message is sent via communication devices carried by store associates or even over the PA system. The reason for the denial (for example, an unknown shopper vs. a known offender) can result in different messaging and resulting actions. In this example, and depending on store policy, associates may simply unlock the fixture and hand the desired merchandise to the unknown shopper while the associate may be required to accompany a known offender to checkout or leave the merchandise at checkout for the known offender on to purchase the way out;

[0111] Video Streaming: Making use of the invention's optional camera, video display, microphone, and speaker, an automatic live-streamed connection may be established between the shopper at the fixture and a store associate (who may or may not actually be in the store); depending on the situation, the associate may remotely command the fixture to unlock or may elect to physically go to the fixture or direct another associate to do so to assist the customer.

[0112] In some situations, there may be a delay before merchandise access is granted. For example, a retailer may elect to deter losses by configuring the system to include a hold-off time for unlocking a fixture between requests from the same shopper or possibly even different shoppers; invoking

a hold-off and its duration may be influenced by the recent pattern of openings as well as the requestor's Trusted Shopper Score. In such a situation, the display provides a countdown or other indication of the holdoff to encourage the shopper to wait.

Applying Trusted Shopper Score to Self-Checkout and Exit Audits

[0113] Retailers are increasingly installing self-checkout stations in their stores at which shoppers scan and pay for their own purchases. Not surprisingly, this process is vulnerable to theft (such as not scanning all items taken by the shopper). Another method of self-checkout enables shoppers to use mobile scanning devices provided by the store or even their own smartphone loaded with a store app to perform checkout as items are selected on the sales floor and placed directly into bags or in a shopping cart. This is even more vulnerable to theft as individuals can easily drop some items into the bags without scanning or even exit the store without scanning any items at all.

[0114] The invention can help reduce this theft activity in three ways:

[0115] Self-Checkout Behavior as in Input to the TSS Score: Most self-checkout systems incorporate various devices, such as analytical video monitoring or product weight measuring devices to detect possible theft activity. Such detections can serve as an input to the trusted shopper score algorithm and could result in triggering a mandatory exit audit of a shopper's receipt against items in their cart, require a "supervised checkout" by a store associate on future visits, or a complete suspension of self-checkout privileges should a pattern emerge.

[0116] Protected Merchandise not Scanned: As previously described, the invention uses various sensors to detect removal of high-value merchandise from secured and, in some cases, unsecured locations. This merchandise removal may be associated with a specific individual. If the removed merchandise is subsequently not scanned by the same individual through a checkout process, the algorithm may characterize that action as an unacceptable behavior that may trigger a receipt audit or lead to revocation of merchandise access and/or self-checkout privileges.

[0117] Selective Exit Auditing: As a further action related to either self-checkout method, detection of removed merchandise not being scanned could result in the trust algorithm triggering a notification or other signaling method such that as the individual approaches a store exit, their purchases may be subjected to an exit audit to verify scanning accuracy.

Advertising, Marketing and Couponing Functions of the Invention

[0118] The invention by its very nature has a "captive audience" in terms of a shopper standing in front of a fixture containing specific merchandise. In addition, leveraging metadata and an individual's classification data, advertising can be customized. This is an ideal environment for targeted advertising to this individual in an attempt to influence their purchase behavior. This advertising capability is integral to the invention and represents "high value real estate" to any potential advertiser.

[0119] The means of advertising is common and not important to the invention. It can be either audio or audio/visual. This can be delivered through the invention's built in audio capability, an external speaker, the invention's built-in display, an auxiliary display located either on the fixture of adjacent to the fixture. Virtually any means of providing promotional messaging can be utilized.

[0120] Selecting the right type of advertisement is core to the invention. The type of advertisement can be selected based on a range of inputs. From very simple, such as what merchandise is contained in the case, to complex, utilizing personal information, metadata and an individual's classification related to the individual requesting access to the locked merchandise. Utilizing this information, the system can deliver significantly more targeted, and therefore more effective advertisements. These advertisements can be brand advertisements, promotions such as advertising a sale or special promotion. They can also be general store ads such as credit card promotions or loyalty membership benefits.

[0121] The following represent certain use cases for the advertising and marketing functions of the invention.

[0122] Merchandise Contained in the Case: In the case where the system is aware of the merchandise contained in the case, either manually entered by the retailer or detected by the merchandise movement sensor, the system can provide intelligent advertising related to and specific to the merchandise contained in the case. This can be product category advertisements or it can be brand specific ads. If the system can detect when a certain SKU is sold out, a similar product can be suggested.

[0123] Personally-Identified Shoppers: In some configurations of the invention, the system knows the shopper "by name". This is the case when the identification methodology is a shopper's loyalty card number, or a retailer's App for example. Once shoppers trade personal identifying information for the convenience of self-service access, the system knows who is accessing the merchandise. This information can be used by the system to provide more targeted advertisements, especially when a shopper's metadata is leveraged. For example, VIP customers can be identified and provided a "thank you" messaging to promote brand loyalty.

[0124] Anonymously-Identified Shoppers: In some configurations, the shopper is identified uniquely but given an anonymous User ID. In this case, the system can keep track of the shopper over time but does not know the shopper's identity. In this case, the system is aware of the products located inside multiple fixture and, based on accumulated shopper purchase behavior identified by which locked cases they have accessed, the system can play targeted advertisements when this shopper is detected dwelling in front of the fixture, or when this shopper is accessing the merchandise.

[0125] Age, Gender, and Mood Identified Shoppers: In some configurations, age, gender and mood data is added to the shopper identification. This information can be used as an input to select appropriate advertising messaging.

[0126] Shopper Dwell: The system is capable of detecting shopper dwell in proximity to the fixture using integrated dwell sensors. In some cases, this sensor cannot uniquely identify the shopper, in other cases, such as with beacons or geofencing via the retailer's app, the system can personally identify the shopper. In this scenario, advertisements can be played enticing shoppers in proximity to the fixture to purchase merchandise contained in the fixture.

[0127] Couponing: As part of the invention's marketing functions, it has the ability to dispense coupons either generally to all individuals, or selectively based on the customer data is has available. Armed with customer data, the system can choose when to dispense a coupon. VIP customers or the Retailer's Loyalty Members may be given a coupon, while non-VIP or non-loyalty customers do not.

[0128] Dynamic Couponing: If the system has the capability to detect which products are chosen from the fixture, a real-time coupon could be presented to the shopper in an attempt to influence their product selection. In a competitive situation, coupons can be dynamically dispensed based on the initial brand selection in the case. For example, if product from Brand A is selected, a coupon for Brand B can be presented. If the shopper initially selects Brand B, no coupon is provided.

[0129] In general, the system can accept inputs from any and all customer specific information, metadata sources, and its own sensor data to make intelligent decisions about advertising and couponing.

Non-Biometric Face Detection

[0130] Due to privacy concerns related to facial recognition and other forms of biometric data collection, some retailers may balk at adopting certain embodiments of the system described above. The alternative method described below avoids these concerns while still delivering many of the system's benefits.

[0131] It is noteworthy that one of the few theft prevention technologies retailers report as effective, even when used over an extended time, is the Public View Monitor (PVM). This ubiquitous device, seemingly installed in every retail store, is simply a video display with an embedded camera such that the viewer sees him or herself in the display. These are generally located at store entrances or at certain high risk merchandise. Some models provide a visual and/or audio indication when motion is detected to promote the inference of active detection. The implication is the video stream is being monitored and/or recorded. However, most PVM's are not even connected externally, much less monitored or recorded. Still, perpetrators are left to contemplate . . . are they watching? Often, the choice is to not take the chance, resulting in theft deterrence.

[0132] The alternative embodiment described here combines the uncertainty instilled by PVM's and with the deterrence of locked merchandise. Much like the invention described earlier using biometric means of uniquely identifying shoppers, shoppers desiring to purchase protected merchandise may do so by simply providing an acceptable image of their face (or other biometric information). This means of accessing locked merchandise is significantly more convenient than calling a store associate. In addition to deterring theft, this approach also averts sales losses typical of secured merchandise and boosts productivity since store staff need not repeatedly go to the merchandise to assist the shopper.

Detailed Description of the Non-Biometric Method

[0133] The following elements typically comprise implementation of this method:

[0134] Video Module: Includes a video display, an audio speaker, at least one (hard or soft) button, and optionally, a wired/wireless external connectivity pro-

visions. Connectivity enables the Video Module to accept information from local sensors, if used, and to communicate with other devices and systems on the store's network, to include communication devices, monitoring stations, and servers. The Video Module is physically small enough to minimize merchandise obstruction and is adjustable to permit aiming the camera to capture the face of users of varying heights, including those in wheelchairs.

[0135] Electronically Controlled Lock: As variety of theft mitigation merchandise fixtures can be outfitted with the invention, the actual locking mechanism will vary. The primary embodiment employs a solenoid or magnetic locking control. Depending on the configuration, a switch or other sensor (which may be a separate device or integral to the locking mechanism) may monitor the fixture to confirm return to its secure state (such as a door in the closed position). Note that some configurations may not use a lock, as described later.

[0136] While reasonable variations of the operating sequence may be implemented, the following is typical.

[0137] First, a person desiring access to the locked merchandise so indicates by pressing a hard or soft button on the touch screen display

[0138] A visual and/or audible message then instructs the user to look directly at the camera typically located above the display. The display presents the customer with an image of the shopper from the embedded camera. The customer aligns their face with a box on the screen, thus providing a perfect "mug shot" of the individual. Upon detection of a face, a tracking box optionally appears around the face and/or a captured image of the face is displayed. No facial recognition occurs; merely detection that someone's human face has been presented to the camera.

[0139] After presence of the face has been detected, a short delay (typically, 3 seconds) occurs to infer processing of the image. Some may assume facial recognition and even identity database lookup is occurring during this time, though that is not the case. Various visual and/or audio cues may be presented during this time. For example, a yellow frame may appear on the perimeter of the display and the words "Processing" displayed and possibly spoken.

[0140] At the end of the delay, the lock is released, allowing the shopper to select the desired merchandise therein.

[0141] During the time merchandise is made available, merchandise removal detection sensors as described above (such as fixture vibration monitoring devices, video, pressure mats, peg hook and pusher dispense counters, and other technologies) detect the quantity of items being removed. If that quantity exceeds the suspicious event threshold also described above, the event is deemed "suspicious" and, depending on system configuration, the system may deploy a local deterrent such as a notification over the system's display, an audible alarm, flashing lights or turn on an external anti-theft deterrent and/or immediately create an alert notification to store staff, local loss prevention personnel, or a remote monitoring station. Such notification is typically accompanied by the image or video captured when merchandise access was granted.

[0142] The image of the person granted access remains on the screen until the transaction terminates, the determination

of which depends on fixture type and system configuration. Three methods define termination:

[0143] Return to Secure State (Door)—This is typically detecting the return of a door on a locked cabinet or a flip door on a merchandise gondola to its closed state. It applies to other securement methods in which releasing the lock permits a person to open a door or other device then, after obtaining the desired merchandise, the door or other device is returned to the closed position, which is detected by the system using a switch or other sensor.

[0144] Controlled Dispense ("Vending" Fixture)—Some secure merchandise fixtures operate similar to a vending machine in which, once access is granted, a person takes an action to dispense one (or possibly more) items. For example, baby formula is a frequently stolen item so is sometimes displayed in a fixture allowing only one item to be removed at a time. Another example are the many methods of limiting access to merchandise typically displayed on peg hooks—from electronically locking peg hooks to "corkscrew" manual twist devices that don't prohibit access but slow it down. In these and many similar applications, an electronic locking mechanism can be fashioned and controlled the invention. by

[0145] Timed Access (Open Fixture)—In some situations, there may be no physical barrier to the merchandise. For example, a vast cosmetics assortment would be difficult or undesirable to obstruct with a physical barrier. In this situation the Video Monitor can be used in conjunction with methods to sense people interacting with merchandise. There exist numerous sensing methods for this including cameras, vibration sensors, infrared virtual fences, pressure mats, and more). Once a face has been detected, the Video Monitor permits access to this merchandise for a defined period of time and/or a quantity of accesses (penetration of hands into the merchandise area), then warns that the interval is ending. Accessing merchandise outside of this time interval can trigger a low level alarm, such as the Video Monitor providing an audio message that access granting is required or, if access persists, a higher alarm level such as notifying loss prevention personnel.

[0146] After termination of access (such as the shopper closing the door), no additional access requests will be processed throughout a configurable hold-off period, such as 30 or 60 seconds. The purpose of this hold-off period is to deter a thief from repeatedly accessing the locked merchandise and, with each access, removing a quantity of items below the suspicious event threshold. The system can be further configured to detect a suspicious pattern of repeated access which, itself, can trigger a suspicious event and/or result in the system going into lockdown mode, disabling further self-service access for a time and/or until reset by local or remote personnel.

[0147] Yet another possible system action under such suspicious circumstances is for the Video Monitor to immediately establish a real time video call with a store associate's communication device (e.g., a smart phone or similar) or to a staffed computer at a remote monitoring station. Should the subsequent conversation warrant, the store associate or monitoring station personnel can command an unlock through the communication device or computer.

[0148] Likewise, should a shopper request access during the hold-off period, the system may be configured to immediately notify a store associate that assistance is needed at that location or, as noted previously, a video call may immediately be established with appropriate personnel who can remotely unlock the merchandise, if warranted. Alternatively, if obstacles to accessing store personnel are anticipated, the video display may provide messaging to encourage a requesting shopper to wait until the hold-off period expires.

[0149] The Video Monitor is capable of providing other benefits when not serving its primary function. For example, promotional information, including videos, regarding the protected merchandise could be displayed. Another available function is simply as a camera with a unique view accessible to the store's local or remote video monitoring station.

In the claims:

1. A system for maximizing sales and minimizing theft in a retail environment, the system comprising:

- (a) providing an electronically activated locking device capable of granting or denying access to a fixture;
- (b) receiving personal identifying information of an individual attempting to access the fixture; and
- (c) instructing the electronically activated locking device to grant or deny access to the fixture, wherein the individual utilizes a mobile device as an interface for entering information into the system or receiving information from the system.

2. The system of claim 1, wherein the personal identifying information of the individual is in a form selected from the group consisting of shopper loyalty information, government issued ID, financial institution cards such as debit cards or credit cards, cell phone numbers, the mobile device which can be identified, near field communication (NFC) technology within the mobile device, a retailer's app comprising login credentials, biometric information, and employee ID cards or employee numbers.

3. The system of claim 1, wherein the locking device further comprises a process for identifying the lock which the individual is attempting to gain access to, the process being at least one selected from the group consisting of: scanning a QR code provided by the locking device which links to a web page; using an app contained on the mobile device capable of scanning a QR code provided by the locking device, using NFC technology on the mobile device, wherein the mobile device is brought into proximity of the locking device and communicates with the locking device, providing at least one biometric sensor which enable the system to identify the individual's biometric information, providing a barcode reader enabling the system to read identification cards which contain similar technology; providing NFC technology which allows the system to read cards or fobs comprising similar technology, wherein the mobile device further comprises Bluetooth capability is brought into proximity of the locking device and communicates with the locking device, providing a radio frequency identification (RFID) sensor enabling the system to read identification cards or fobs.

4. The system of claim 1, wherein the interface of the mobile device is a user interface which is used for entering information into the system or receiving information from the system.

5. The system of claim 1, wherein the individual utilizes an app on the mobile device as the interface for entering information into the system or receiving information from the system

6. The system of claim 1, wherein multiple fixtures can be controlled from a single interface.

7. The system of claim 6, wherein the individual enters a locking device identifier into the single interface to indicate which locking device is accessed.

8. The system of claim 6, wherein the system unlocks all locking devices in a local area once the individual is authenticated using the personal identifying information.

9. The system of claim 6, wherein the system re-locks the locking device once the fixture is opened or after a programmable delay.

10. The system of claim 1, wherein at least one camera captures an image or video of the individual interacting with the merchandise contained within the fixture or during an interaction with the fixture onto which the locking device is assembled.

11. The system of claim 10, wherein the image or video of the individual interacting with the fixture or the merchandise is stored in at least one database.

12. The system of claim 10, wherein video of the individual interacting with the fixture or the merchandise is integrated with a video management system, wherein the video management system is capable of bookmarking video from at least one additional camera.

13. The system of claim 1, wherein the locking device is computer controlled.

14. The system of claim 1, wherein the system can automatically re-lock following an access enabling event.

15. The system of claim 1, wherein the system transmits the personal identifying information and other data associated with the individual's interaction with the locking device to at least one database.

16. The system of claim 15, wherein the personal identifying information of the individual is correlated with personal identifying information contained in at least one external database.

17. The system of claim 16, wherein the information from the at least one database and the at least one external database is used to determine if the individual will be granted access.

18. The system of claim 16, wherein the information from the least one database and the at least one external database contains information about the individual's past behaviors.

19. The system of claim 1, wherein the device comprises at least one sensor.

20. The system of claim 19, wherein the system calculates a time duration during which the device is in an open state after being accessed by the individual.

21. The system of claim 16, wherein the device alarms if the time duration of an open condition exceeds a predetermined threshold.

22. The system of claim 15, wherein the information from the at least one database contains information about the individual including at least one from the group consisting of personal identifying information and shopping behavior factors of the individual.

23. The system of claim 16, wherein the information from the at least one external database is at least one selected from the group consisting of personal identifying information, known offenders, case management data, purchase history,

loyalty card membership, app login credentials, authorized vendor personnel, active employees, terminated employees, and VIP customers.

24. The system of claim **15**, wherein the system further comprises a switch or other sensor, which may be separate from or integrated within the locking device.

25. The system of claim **24**, wherein the system can differentiate normal device interactions from suspicious interactions.

26. The system of claim **25**, wherein the system contains thresholds for each interaction behavior delineating suspicious behaviors from normal behaviors.

27. The system of claim **26**, wherein the thresholds are established using statistical methods applied to historical behaviors.

28. The system of claim **26**, wherein the thresholds are established manually.

29. The system of claim **26**, wherein the system locally alarms when a behavior threshold is crossed indicating suspicious behavior.

30. The system of claim **26**, wherein the system can trigger an alarm or issue an alert through a PA system or other communication device when a behavior threshold is crossed indicating suspicious behavior.

31. The system of claim **26**, wherein if the threshold for the suspicious behavior is crossed, the individual's subsequent access to the fixture is denied for a set period of time.

32. The system of claim **17**, wherein the system denies access to the individual if that individual's personal identifying information is correlated to at least one suspicious event from the at least one database or the at least one external database.

33. The system of claim **23**, wherein at least one algorithm determines whether the system grants or denies access to the individual.

34. The system of claim **22**, wherein at least one algorithm determines whether the system grants or denies access to the individual.

35. The system of claim **1**, wherein the system further comprises a feature which allows the individual to request assistance in accessing the fixture.

36. The system of claim **19**, wherein the at least one sensor identifies when the locking device is being accessed by the individual.

37. The system of claim **24**, wherein the system monitors the fixture to confirm a return to its secure state.

38. The system of claim **24**, wherein the locking device can identify whether it is in a locked or unlocked state.

39. A system for maximizing sales and minimizing theft in a retail environment, the system comprising:

- (a) providing an electronically activated locking device capable of granting or denying access to a fixture;
- (b) receiving personal identifying information of an individual attempting to access the fixture; and
- (c) instructing the electronically activated locking device to grant or deny access to the fixture; and
- (d) utilizing at least one biometric identification method for identifying the individual.

40. A system for maximizing sales and minimizing theft in a retail environment, the system comprising:

- (a) providing an electronically activated locking device which is capable of granting or denying access to a fixture;
- (b) receiving personal identifying information of an individual attempting to access the fixture; and
- (c) instructing the electronically activated locking device to grant or deny access to the fixture.

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